

is claimed to be the world's first self-loading digital manifest. The IQtag is powered by the Nordic Semiconductor nRF52832 SoC based 'I-SYST Blyst Nano' module. The Bluetooth LE-enabled beacon is a durable and



lightweight wearable solution that can transmit crucial information, such as names

and weights of crew members, cargo, and passengers. The data is sent to the onboard app via Bluetooth LE connectivity, providing real-time visibility of individuals and items onboard the transport. The IQtag system streamlines communication, reduces reliance on radio calls, eliminates manual calculations, and offers accurate weight and balance calculations for pilots and dispatchers. The solution can enhance aviation safety and alleviate pilot workload in helicopter operations for services like air ambulances, search and rescue missions, oil and gas transportation, tourism, executive/VIP transport, and firefighting.

IQonboard

<https://iqonboard.com>

Capacitive fingerprint recognition module

Rajguru Electronics has launched the R503 capacitive fingerprint recognition module with a new circular design and two-colour ring indicator LED control.



It seamlessly integrates into access control systems, attendance trackers,

safety deposit boxes, and more. The module has an integrated image collecting and algorithm chip, offering flexibility for different finger conditions with a high recognition rate. It supports serial communication with a microcontroller unit at 3.3V or 5V. It uses series data exchange format for communication with a computer, com-

patible with UART and USB modes. A USB interface is recommended for PCs to improve exchange speed in fingerprint scanning devices.

Rajguru Electronics

<https://www.rajguruelectronics.com>

BOARDS AND MODULES

AC/DC power supply

COSEL HCA3500TF is a 3-phase/3-wire 3,500W AC/DC enclosed power supply optimised for such industrial



equipment as high-frequency amplifiers, laser processing machines, and

robotics. The versatility of the power supply allows it to be installed worldwide and powered by grids ranging from 200 to 400V. It supports 3-phase, 3-wire Delta or Star installations, making it suitable for system architects aiming to improve load-balancing and reduce single-phase peak loading.

The power supply caters to a diverse range of applications, including measurement and analysis equipment, machine tools, and semiconductor manufacturing. It features an optimised, digitally controlled switching topology and employs SiC MOSFETs and diodes for maximum efficiency and thermal robustness. The HCA3500 also features such protection features as inrush current limiting, overcurrent and overvoltage protection, and thermal protection.

COSEL Co, Ltd

<https://en.cosel.co.jp>

5W DC/DC converters

Recom REC5K AW series 5W DC/DC converters come in miniature packages with a compact 25.4mm × 25.4mm footprint and a 4:1 input range. These cost-effective converters provide a regulated supply of 5V output at 1A, accommodating an input range of 18 to 36V DC and an operating temperature range of -40 to +110°C. Moreover, these converters offer

various safety features, such as the short-circuit protection and an input undervoltage lockout. With a standby current draw of only 3mA and the



ability to operate at high altitudes and in pollution degree 2 (PD2) environments, the converters

are suitable for use in harsh environments. The REC5K AW series is certified for 'basic' grade isolation and meets EMC standard Class B for input conducted noise with a simple input filter. The converters will enable power electronics engineers to build cost-effective and robust power supplies that can be used for critical medical and industrial applications that are suitable for deployment even in higher altitudes.

Recom Power

<https://recom-power.com>

1W DC/DC converter

Traco Power's TRI 1 series high-isolation DC/DC converters feature 1-watt power and 480V AC isolation voltage. They offer a reinforced isolation system capable of withstanding high



test voltages and working voltages of 480V AC

RMS. Moreover, the power converters feature an efficiency rate of 84% and are suitable for use in extreme temperature conditions ranging from -40 to +85°C without derating. The TRI 1 series is designed for low-power applications in industrial, transportation, and instrumentation sectors. The compact converters come in SIP-8 and SMD-14 packages and offer a ±10% input voltage range with precisely regulated and isolated output voltages. They accept input voltages from 4.5 to 26.4V DC and deliver 5V DC, 12V DC, or 15V DC outputs. The 1-watt DC/DC converters also features reinforced I/O isolation, ultra-high isolation peak voltage, wide operating temperature